NCP shadow reachability: exact Lean equivalence

2026-10-21; THM-COMMUTATOR-KERNEL1 Lemma 2(b); TASK-20260822-3dd086

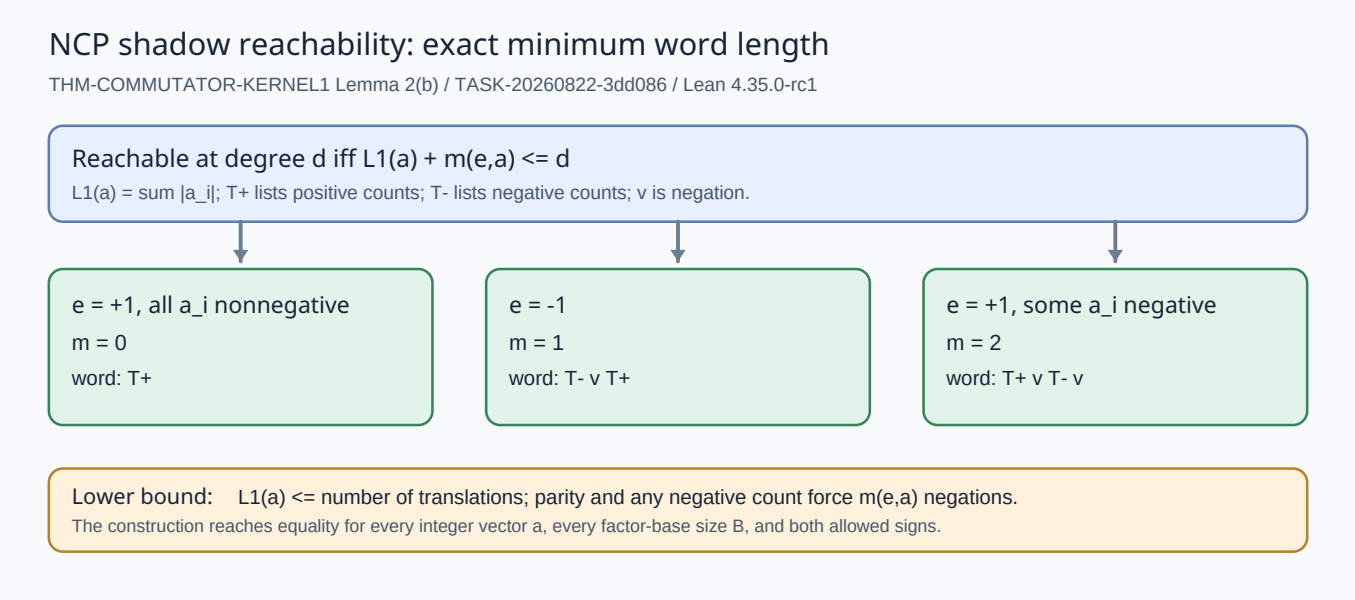
Scope. This proof characterizes the minimum word length needed to represent a signed-count shadow of translation and negation letters. It uses the same right-of-occurrence sign convention as the earlier affine-action proof. It makes no search-cost or elliptic-curve performance claim.

Statement and construction

For $a \in \mathbb{Z}^B$, write $\|a\|_1 = \sum_i |a_i|$. The sign e belongs to $\{+1, -1\}$. The penalty $m(e, a)$ is 0 for $e = +1$ with every $a_i \geq 0$, 1 for $e = -1$, and 2 for $e = +1$ with some $a_i < 0$. Lean proves, for every B, a, and degree bound d,

$$(\exists w,\ |w| \leq d\ \wedge\ \text{shadow}(w) = (e, a)) \quad \Longleftrightarrow \quad \|a\|_1 + m(e, a) \leq d.$$

The lower bound charges one letter per unit of signed-count weight and the necessary negations. For the upper bound, T^+ and T^- list the positive and negative coordinate counts. The three exact constructions are T^+, $T^-\nu T^+$, and $T^+\nu T^-\nu$.



Validation and limits

Check	Observed result
Pinned Lean target	Machine verified on Lean 4.35.0-rc1; project audit passed 498 theorem declarations using standard axioms only. Independent semantic review remains pending.
Direct shadow crosscheck	28 (B, d) cases, 9,190 exhaustive word evaluations, zero missing or extra shadows for $B = 0..3$, $d = 0..6$.
Complete target corpus	65 machine verified; zero failed; zero infrastructure failures; one theorem file absent (NCP commutator-ideal refutation).

The bounded crosscheck compares unreduced shadow pairs. The formal theorem holds at every size, but it does not prove the path-algebra kernel ideal, elliptic-curve code refinement, NC search cost, or ECDLP performance. No new timing datum was produced, so canonical numerical dashboards are unchanged.